Qualification Pack

Retail Outlet Roof Top Solar Installer- Oil &Gas

QP Code: HYC/Q3103

Version: 1.0

NSQF Level: 4

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HYC/Q3103: Retail Outlet Roof Top Solar Installer-Oil &Gas

Brief Job Description

A roof top solar panel installer is responsible for installing and servicing photovoltaic (PV) solar panels on the roofs of the retail outlets / CNG gas stations ensuring that the methods and tools adhere to safety regulations and practices. His/her job is to ensure that the installed solar panels efficiently convert solar energy into renewable power to be utilised at the above installations. Installers are also responsible for maintaining the solar panels and ensuring that the overall design and the wiring systems are safe and efficient. Most of your work as a solar panel installer will be outdoors and at height.

Personal Attributes

The individual should have a knowledge about the electrical systems and able to perform mathematical calculations, besides having ability to give attention to detail, technical & analytical aptitude, adaptability, teamwork and a strong commitment to safety. The person should have the ability to work individually / team round the clock and must be able to stay calm in stressful situations. This is a hands-on position that requires physical dexterity and attention to detail.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. HYC/N3115: Introduction to Solar Technologies and Role & Responsibilities of a Solar Panel Installer at a Retail Outlet
2. HYC/N3116: Assembling and Commissioning of Rooftop Solar PV System at a Fuel Station
3. HYC/N3117: Maintaining Solar PV System and Ensuring Safety at a Fuel Station
4. HYC/N9301: Working Effectively in a team
5. HYC/N9302: Maintain health, safety and security procedures
6. DGT/VSQ/N0102: Employability Skills (60 Hours)

Qualification Pack (QP) Parameters

Sector	Hydrocarbon
Sub-Sector	Downstream
Occupation	Retail Distribution

Qualification Pack

Country	India
NSQF Level	4
Credits	16
Aligned to NCO/ISCO/ISIC Code	NCO/2015-5245.010
Minimum Educational Qualification & Experience	12th grade Pass (Science) with 1-2 Years of experience OR Completed 3 year diploma after 10th (in relevant branch) OR 10th grade pass with 3 Years of experience
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	29/05/2027
NSQC Approval Date	30/05/2024
Version	1.0
Reference code on NQR	QG-04-HY-02621-2024-V1-HSSC
NQR Version	1.0

HYC/N3115: Introduction to Solar Technologies and Role & Responsibilities of a Solar Panel Installer at a Retail Outlet

Description

This unit covers fundamental aspects related to electricity and provide an overview of different solar technologies, their basic principles & applications, as well as types of PV systems and their components.

Scope

The scope covers the following :

- This unit/task covers the following
- Basics about Electricity
- Different Solar technologies
- Role & responsibilities of a roof top solar installer

Elements and Performance Criteria

Basics of Electricity

To be competent, the user/individual on the job must be able to:

PC1. Understand the global and Indian energy scenario
PC2. Define electricity and Know what is current, voltage and its origin.
PC3. Know what is resistance
PC4. Use Ohm's Law to calculate current, voltage, and resistance
PC5. Differentiate electric power from energy
PC6. Differentiate between direct current (DC) and alternate current (AC)
PC7. Understand the working of batteries
PC8. Understand working of a transformer
PC9. Explain the difference between conductors and insulators
PC10. Calculate equivalent resistance for series, parallel, or series-parallel circuits
PC11. Calculate voltage drop across a resistor
PC12. Calculate power given other basic values
PC13. Identify factors that determine the strength and polarity of a current-carrying coil's magnetic field
PC14. Determine peak, instantaneous, and effective values of an AC sine wave
PC15. Identify factors that effect inductive reactance and capacitive reactance in an AC circuit
PC16. Calculate total impedance of an AC circuit
PC17. Explain the difference between real power and apparent power in an AC circuit
PC18. Calculate primary and secondary voltages of single-phase and three-phase transformers
PC19. Calculate kVA of a transformer

Different Solar Technologies

To be competent, the user/individual on the job must be able to:

PC20. Identify three primary technologies for harnessing solar energy
PC21. Understand photovoltaic (PV) effect for direct conversion of light to electricity
PC22. Understand concentrating solar power (CSP) and its working
PC23. Know how solar thermal energy can be used to provide hot water and air conditioning.
PC24. Identify advantages of Silicon solar cells
PC25. Identify fundamental difference between the Dye Sensitised Solar cells (DSSC) and the conventional solar cells
PC26. Know about Thin Film Solar Cells and their advantages.
PC27. Understand III-V types of solar cells
PC28. Know about next-generation solar cell technologies.
PC29. Explain efficiency of a solar cell
PC30. Identify if the current produced by Solar panels is DC or AC?
PC31. Know how to convert DC into AC using Inverters.
PC32. Identify how to store the DC energy into batteries
PC33. Understand about concentric solar power (CSP)
PC34. List down different types of CSP technologies.
PC35. Identify differences between PV and CSP technologies.
PC36. Understand how PV Solar Panels /Modules are formed from solar cells
PC37. Understand how PV panels can be connected in groups to form a PV array.
PC38. explain the string and array definition of solar module
PC39. discuss the string voltage and current
PC40.
- discuss the Open circuit voltage and short circuit current and measure the open circuit voltage
- and short circuit current of solar module

PC41.
- discuss the Vmp (voltage at maximum power) and Imp(Current at maximum power) of Solar
- module and measure the Vmp and Imp of solar module

PC42. discuss the standard test conditions of Solar module
PC43. discuss the different power rating of solar module and its importance
PC44. demonstrate different type of solar cell and solar module
PC45. Explain and understand DNI, GHI and Diffused Irradiance & Irradiation.
PC46. Understand the movement of the sun and its effect on the performance of the plant
PC47. Know the importance and working of the tracking systems.

Roles & Responsibilities of a Roof Top Solar Installer at Retail outlet

To be competent, the user/individual on the job must be able to:

PC48. Assess Retail Outlet locations to ensure they are suitable for solar panels
PC49. Measuring up fuel station areas for installation
PC50. Mount and setting solar panels at Retail Outlet
PC51. Manage telecommunications, data and CCTV equipment.
PC52. Do all electrical connections from AC to DC.
PC53. Install framing, wiring (including wire management) and PV modules
PC54. Route all DC wire and splice connections.
PC55. Install PVC pipe underground, running to panels.

PC56. Design review, QC, and site audit
PC57. Disconnect high risers and rebuild them as a new connector.
PC58. Build solar panels, drill holes, dig trenches for electrical.
PC59. Install relay switches, on and off switches, and disconnect switches.
PC60. Cut, thread and fit plumbing according to specifications for connecting circulation system.
PC61. Climb ladders with heavy equipment and solar panels, to be installed on Fuel station roofs.
PC62. Making adjustments to building surfaces so panels fit and are adequately supported
PC63. Install equipment such as batteries, rectifiers, PBD's, BDFB's and associate infrastructure used in telecommunications networks.
PC64. Complete the circuits between the solar panels and run the PV wires through the racking rows.
PC65. Used knowledge of electrical construction, roofing construction, general construction and/or relevant codes.
PC66. Assume all safety precautions along with all state and OSHA (Occupational Safety and Health Administration) requirements for the job.
PC67. Install batteries and rectifiers in Telecom environments.
PC68. Perform basic installation skills of plumbing, carpentry, electrical in the installation of solar hot water systems.
PC69. Wire and connect solar panels to each other and wire solar panels to a DC/AC electrical converter.
PC70. Ground the electrical equipment and check wiring
PC71. Have a sound understanding of the relevant codes that relate to PV installations.
PC72. Assure everyone are safe and wearing proper PPE
PC73. Train and run crews in solar installation
PC74. Use variety of technical equipment and tools
PC75. Performing maintenance checks and fixing solar panels

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Basics of electricity and electrical connections
KU2. Basic Mathematical calculations
KU3. Codes and regulations for solar installations in India
KU4. Usage of basic tools and equipment used in electrical industry
KU5. Emergency response and safety skills
KU6. different types of earthing and lightning protection
KU7. reporting protocol and documentation required
KU8. methods of reducing electrical consumptions
KU9. types of meters, bidirectional and unidirectional for measuring import and export of electricity
KU10. methods of reducing electrical consumptions

KU11. renewable energy sources which can be deployed at the workplace
KU12. gender and its concepts such as gender roles, gender spectrum, gender as an identity
KU13. the concept of Mission Karmayogi
KU14. the concept of Swachhata Hi Seva

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. communicate effectively and share information efficiently
GS2. Effective use of computers (MS office tools)
GS3. Presentation skills -oral and written.
GS4. Effectively use read and interpret the data
GS5. read instructions, guidelines/procedures and reports
GS6. analyze the parameters and output
GS7. communicate effectively with team members.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Basics of Electricity	**19**	**38**	**-**	**-**
PC1. Understand the global and Indian energy scenario	1	2	-	-
PC2. Define electricity and Know what is current, voltage and its origin.	1	2	-	-
PC3. Know what is resistance	1	2	-	-
PC4. Use Ohm's Law to calculate current, voltage, and resistance	1	2	-	-
PC5. Differentiate electric power from energy	1	2	-	-
PC6. Differentiate between direct current (DC) and alternate current (AC)	1	2	-	-
PC7. Understand the working of batteries	1	2	-	-
PC8. Understand working of a transformer	1	2	-	-
PC9. Explain the difference between conductors and insulators	1	2	-	-
PC10. Calculate equivalent resistance for series, parallel, or series-parallel circuits	1	2	-	-
PC11. Calculate voltage drop across a resistor	1	2	-	-
PC12. Calculate power given other basic values	1	2	-	-
PC13. Identify factors that determine the strength and polarity of a current-carrying coil's magnetic field	1	2	-	-
PC14. Determine peak, instantaneous, and effective values of an AC sine wave	1	2	-	-
PC15. Identify factors that effect inductive reactance and capacitive reactance in an AC circuit	1	2	-	-
PC16. Calculate total impedance of an AC circuit	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC17. Explain the difference between real power and apparent power in an AC circuit	1	2	-	-
PC18. Calculate primary and secondary voltages of single-phase and three-phase transformers	1	2	-	-
PC19. Calculate kVA of a transformer	1	2	-	-
Different Solar Technologies	**28**	**56**	**-**	**-**
PC20. Identify three primary technologies for harnessing solar energy	1	2	-	-
PC21. Understand photovoltaic (PV) effect for direct conversion of light to electricity	1	2	-	-
PC22. Understand concentrating solar power (CSP) and its working	1	2	-	-
PC23. Know how solar thermal energy can be used to provide hot water and air conditioning.	1	2	-	-
PC24. Identify advantages of Silicon solar cells	1	2	-	-
PC25. Identify fundamental difference between the Dye Sensitised Solar cells (DSSC) and the conventional solar cells	1	2	-	-
PC26. Know about Thin Film Solar Cells and their advantages.	1	2	-	-
PC27. Understand III-V types of solar cells	1	2	-	-
PC28. Know about next-generation solar cell technologies.	1	2	-	-
PC29. Explain efficiency of a solar cell	1	2	-	-
PC30. Identify if the current produced by Solar panels is DC or AC?	1	2	-	-
PC31. Know how to convert DC into AC using Inverters.	1	2	-	-
PC32. Identify how to store the DC energy into batteries	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC33. Understand about concentric solar power (CSP)	1	2	-	-
PC34. List down different types of CSP technologies.	1	2	-	-
PC35. Identify differences between PV and CSP technologies.	1	2	-	-
PC36. Understand how PV Solar Panels /Modules are formed from solar cells	1	2	-	-
PC37. Understand how PV panels can be connected in groups to form a PV array.	1	2	-	-
PC38. explain the string and array definition of solar module	1	2	-	-
PC39. discuss the string voltage and current	1	2	-	-
PC40. • discuss the Open circuit voltage and short circuit current and measure the open circuit voltage • and short circuit current of solar module	1	2	-	-
PC41. • discuss the Vmp (voltage at maximum power) and Imp(Current at maximum power) of Solar • module and measure the Vmp and Imp of solar module	1	2	-	-
PC42. discuss the standard test conditions of Solar module	1	2	-	-
PC43. discuss the different power rating of solar module and its importance	1	2	-	-
PC44. demonstrate different type of solar cell and solar module	1	2	-	-
PC45. Explain and understand DNI, GHI and Diffused Irradiance & Irradiation.	1	2	-	-
PC46. Understand the movement of the sun and its effect on the performance of the plant	1	2	-	-
PC47. Know the importance and working of the tracking systems.	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Roles & Responsibilities of a Roof Top Solar Installer at Retail outlet	**28**	**56**	**-**	**-**
PC48. Assess Retail Outlet locations to ensure they are suitable for solar panels	1	2	-	-
PC49. Measuring up fuel station areas for installation	1	2	-	-
PC50. Mount and setting solar panels at Retail Outlet	1	2	-	-
PC51. Manage telecommunications, data and CCTV equipment.	1	2	-	-
PC52. Do all electrical connections from AC to DC.	1	2	-	-
PC53. Install framing, wiring (including wire management) and PV modules	1	2	-	-
PC54. Route all DC wire and splice connections.	1	2	-	-
PC55. Install PVC pipe underground, running to panels.	1	2	-	-
PC56. Design review, QC, and site audit	1	2	-	-
PC57. Disconnect high risers and rebuild them as a new connector.	1	2	-	-
PC58. Build solar panels, drill holes, dig trenches for electrical.	1	2	-	-
PC59. Install relay switches, on and off switches, and disconnect switches.	1	2	-	-
PC60. Cut, thread and fit plumbing according to specifications for connecting circulation system.	1	2	-	-
PC61. Climb ladders with heavy equipment and solar panels, to be installed on Fuel station roofs.	1	2	-	-
PC62. Making adjustments to building surfaces so panels fit and are adequately supported	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC63. Install equipment such as batteries, rectifiers, PBD's, BDFB's and associate infrastructure used in telecommunications networks.	1	2	-	-
PC64. Complete the circuits between the solar panels and run the PV wires through the racking rows.	1	2	-	-
PC65. Used knowledge of electrical construction, roofing construction, general construction and/or relevant codes.	1	2	-	-
PC66. Assume all safety precautions along with all state and OSHA (Occupational Safety and Health Administration) requirements for the job.	1	2	-	-
PC67. Install batteries and rectifiers in Telecom environments.	1	2	-	-
PC68. Perform basic installation skills of plumbing, carpentry, electrical in the installation of solar hot water systems.	1	2	-	-
PC69. Wire and connect solar panels to each other and wire solar panels to a DC/AC electrical converter.	1	2	-	-
PC70. Ground the electrical equipment and check wiring	1	2	-	-
PC71. Have a sound understanding of the relevant codes that relate to PV installations.	1	2	-	-
PC72. Assure everyone are safe and wearing proper PPE	1	2	-	-
PC73. Train and run crews in solar installation	1	2	-	-
PC74. Use variety of technical equipment and tools	1	2	-	-
PC75. Performing maintenance checks and fixing solar panels	1	2	-	-
NOS Total	**75**	**150**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N3115
NOS Name	Introduction to Solar Technologies and Role & Responsibilities of a Solar Panel Installer at a Retail Outlet
Sector	Hydrocarbon
Sub-Sector	
Occupation	Retail Distribution
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

HYC/N3116: Assembling and Commissioning of Rooftop Solar PV System at a Fuel Station

Description

This unit covers the description of various components of a rooftop PVsystems, as well as how to design the PV system, procedure for installation and commissioning of the PV system.

Scope

The scope covers the following :

- This unit/task covers the following
- PV System Components
- PV System Design, Installation & Commissioning

Elements and Performance Criteria

PV System Components

To be competent, the user/individual on the job must be able to:

PC1. Understand basic design principles and components of PV systems.

PC2. Assemble the PV panels/array as per the requirements by electrically connecting the panels

PC3. Know about the direct-current systems

PC4. Understand the function of Combiner box

PC5. Identify the battery charge controllers to regulates the flow of electricity from the PV modules to the battery and the load

PC6. discuss the type of charge controller and advantage and disadvantage

PC7. Understand the function of battery to store electricity when sun is not shining

PC8. Know about the alternate-current system

PC9. Know about the PV inverters for converting DC into AC for loads requiring alternating current,

PC10. match the power quality required by loads with the power quality produced by the inverter

PC11. Install the electrical utility meter to record the power consumption

PC12. Identify Overcurrent and surge protection devices

PC13. Gauge the strength and design of PV Module Mounting iron Structure

PC14. Discuss the role of transformer in increasing the voltage as per requirements

PC15. Identity DC and AC switches

PC16. List down the cables, connectors & accessories required for the installation

PC17. Identity conduits, fuses, safety disconnects etc

PC18. Know the grounding equipments

PC19. Understand BIS codes, standards & certification of all the PV system components

PV System Design, Installation & Commissioning

To be competent, the user/individual on the job must be able to:

PC20. Estimate the load requirements of the client and analyse the same

PC21. Carry out the site survey and prepare the report
PC22. prepare a layout sketch of the site and include into the report
PC23. identify and highlight the major obstacle on site
PC24. Suggest a suitable site based on the sun light availability and shadow analysis
PC25.
- discuss technical specification of existing electrical system (Breaker capacity, spare
- feeder detail, etc)

PC26. Identify number of PV panels required
PC27. Estimate size of the battery bank
PC28. Prepare drawings for the roof top PV system and discuss with the client
PC29. Design the complete PV system based on above
PC30. Ensure that all the PV system components comply to BIS codes and standards, and necessary certificates are available
PC31. Estimate the total area requirement keeping safety aspects in mind
PC32. Prepare cost estimate for the entire system
PC33. Understand the function of a PV system monitoring
PC34. Follow step by step instructions /guidelines for installation of solar panels
PC35. Install the mounting framework as per the design (height, tilt angle etc)
PC36. To Put the solar panels secured to the mounting structure
PC37. Carry out the electrical wiring.
PC38. Connect the solar inverter to the system
PC39. Connect the solar inverter to the solar battery
PC40. Link the solar inverter to the power grid
PC41. Complete all the electrical cabling & connections
PC42. Follow the commissioning procedure and check list
PC43. Check the system thoroughly before the commissioning start
PC44. Ensure that all disconnects are in OFF position during the final checkout.
PC45. Check that Installation matches the design documentation.
PC46. Ensure that system components have compliance with the national codes & standards
PC47. Check that terminals connections and screws are securely tightened.
PC48. Ensure that equipment is securely mounted (such as modules, racking, inverters, panels, and disconnects and so on)
PC49. Check that warning signs and labels are posted appropriately
PC50. Ensure that job site is clean and all tools etc removed from the site before commissioning.
PC51. Understand that commissioning procedure starts at the array and ends at the point of connection.
PC52. To connect power sources to systems-connecting PV module wire runs
PC53. Test DC voltage and polarity
PC54. Test AC voltage at inverter output
PC55. Start the PV system and show to the customer.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. different types of earthing and lightning protection
KU2. reporting protocol and documentation required
KU3. methods of reducing electrical consumptions
KU4. types of meters, bidirectional and unidirectional for measuring import and export of electricity
KU5. methods of reducing electrical consumptions
KU6. renewable energy sources which can be deployed at the workplace
KU7. gender and its concepts such as gender roles, gender spectrum, gender as an identity
KU8. company's code of conduct
KU9. importance of individual's role in the work flow
KU10. organisational culture
KU11. effect of harmonic current injection, flicker and DC injection into AC grid on quality of power
KU12. tools & tackles required for inspection
KU13. do's and don'ts of DC wiring and installation of other electrical components
KU14. testing and commissioning activities and its interpretation - visual inspection, continuity of wiring, earthing, polarity check, insulation, voltage drop etc
KU15. typical faults, their causes and resolution for all system components
KU16. basics of electricity and prevalent energy efficient devices
KU17. ways to recognize common electrical problems
KU18. common practices of conserving electricity
KU19. the concept of Mission Karmayogi
KU20. the concept of Swachhata Hi Seva

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. communicate effectively and share the information efficiently
GS2. Effective use of computers (MS office tools)
GS3. Presentation skills -oral and written.
GS4. Effective use read and interpret the data
GS5. read instructions, guidelines/procedures and reports
GS6. analyze the parameters and output
GS7. communicate effectively with team members.
GS8. plan and organize work to meet deadlines

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PV System Components	**19**	**38**	**-**	**-**
PC1. Understand basic design principles and components of PV systems.	1	2	-	-
PC2. Assemble the PV panels/array as per the requirements by electrically connecting the panels	1	2	-	-
PC3. Know about the direct-current systems	1	2	-	-
PC4. Understand the function of Combiner box	1	2	-	-
PC5. Identify the battery charge controllers to regulates the flow of electricity from the PV modules to the battery and the load	1	2	-	-
PC6. discuss the type of charge controller and advantage and disadvantage	1	2	-	-
PC7. Understand the function of battery to store electricity when sun is not shining	1	2	-	-
PC8. Know about the alternate-current system	1	2	-	-
PC9. Know about the PV inverters for converting DC into AC for loads requiring alternating current,	1	2	-	-
PC10. match the power quality required by loads with the power quality produced by the inverter	1	2	-	-
PC11. Install the electrical utility meter to record the power consumption	1	2	-	-
PC12. Identify Overcurrent and surge protection devices	1	2	-	-
PC13. Gauge the strength and design of PV Module Mounting iron Structure	1	2	-	-
PC14. Discuss the role of transformer in increasing the voltage as per requirements	1	2	-	-
PC15. Identity DC and AC switches	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. List down the cables, connectors & accessories required for the installation	1	2	-	-
PC17. Identity conduits, fuses, safety disconnects etc	1	2	-	-
PC18. Know the grounding equipments	1	2	-	-
PC19. Understand BIS codes, standards & certification of all the PV system components	1	2	-	-
PV System Design, Installation & Commissioning	**36**	**72**	**-**	**-**
PC20. Estimate the load requirements of the client and analyse the same	1	2	-	-
PC21. Carry out the site survey and prepare the report	1	2	-	-
PC22. prepare a layout sketch of the site and include into the report	1	2	-	-
PC23. identify and highlight the major obstacle on site	1	2	-	-
PC24. Suggest a suitable site based on the sun light availability and shadow analysis	1	2	-	-
PC25. • discuss technical specification of existing electrical system (Breaker capacity, spare • feeder detail, etc)	1	2	-	-
PC26. Identify number of PV panels required	1	2	-	-
PC27. Estimate size of the battery bank	1	2	-	-
PC28. Prepare drawings for the roof top PV system and discuss with the client	1	2	-	-
PC29. Design the complete PV system based on above	1	2	-	-
PC30. Ensure that all the PV system components comply to BIS codes and standards, and necessary certificates are available	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC31. Estimate the total area requirement keeping safety aspects in mind	1	2	-	-
PC32. Prepare cost estimate for the entire system	1	2	-	-
PC33. Understand the function of a PV system monitoring	1	2	-	-
PC34. Follow step by step instructions /guidelines for installation of solar panels	1	2	-	-
PC35. Install the mounting framework as per the design (height, tilt angle etc)	1	2	-	-
PC36. To Put the solar panels secured to the mounting structure	1	2	-	-
PC37. Carry out the electrical wiring.	1	2	-	-
PC38. Connect the solar inverter to the system	1	2	-	-
PC39. Connect the solar inverter to the solar battery	1	2	-	-
PC40. Link the solar inverter to the power grid	1	2	-	-
PC41. Complete all the electrical cabling & connections	1	2	-	-
PC42. Follow the commissioning procedure and check list	1	2	-	-
PC43. Check the system thoroughly before the commissioning start	1	2	-	-
PC44. Ensure that all disconnects are in OFF position during the final checkout.	1	2	-	-
PC45. Check that Installation matches the design documentation.	1	2	-	-
PC46. Ensure that system components have compliance with the national codes & standards	1	2	-	-
PC47. Check that terminals connections and screws are securely tightened.	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC48. Ensure that equipment is securely mounted (such as modules, racking, inverters, panels, and disconnects and so on)	1	2	-	-
PC49. Check that warning signs and labels are posted appropriately	1	2	-	-
PC50. Ensure that job site is clean and all tools etc removed from the site before commissioning.	1	2	-	-
PC51. Understand that commissioning procedure starts at the array and ends at the point of connection.	1	2	-	-
PC52. To connect power sources to systems- connecting PV module wire runs	1	2	-	-
PC53. Test DC voltage and polarity	1	2	-	-
PC54. Test AC voltage at inverter output	1	2	-	-
PC55. Start the PV system and show to the customer.	1	2	-	-
NOS Total	**55**	**110**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N3116
NOS Name	Assembling and Commissioning of Rooftop Solar PV System at a Fuel Station
Sector	Hydrocarbon
Sub-Sector	
Occupation	Retail Distribution
NSQF Level	4
Credits	3.5
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

HYC/N3117: Maintaining Solar PV System and Ensuring Safety at a Fuel Station

Description

This unit covers the maintenance part of the Fuel Station roof top solar PV system, including preventive and regular maintenance. The unit will also cover various safety precautions & procedures to be followed by a technician.

Scope

The scope covers the following :

- This unit/task covers the following
- Monitoring & maintaining the fuel station Solar rooftop installation
- Solar PV Safety

Elements and Performance Criteria

Monitoring & Maintaining the Fuel Station Solar rooftop installation

To be competent, the user/individual on the job must be able to:

PC1. Understand different strategies adopted for maintenance of a solar roof top PV system- Routine maintenance, Preventative maintenance, Corrective or Reactive maintenance, and Condition-based maintenance.

PC2. Understand the objective of preventive maintenance

PC3. Understand the objective of a PV system monitoring on regular basis

PC4. Track energy production and usage in real-time

PC5. Identify dips in output or problems with individual panels, batteries, or other system components.

PC6. Understand the sensors and communication devices attached to solar arrays

PC7. Read, record, & store the voltage, current, and power output in real time

PC8. Carry out routine inspection and servicing of equipment to prevent breakdowns and unnecessary production losses.

PC9. To regularly check basic maintenance elements such as modules, inverters, charge controllers, and batteries.

PC10. Carry out a thorough inspection of PV modules visually to look for any issues

PC11. Look for physical damage to frame and glass etc and repair the same

PC12. Identify any delamination or change in color of the module's outer layer due to separation of bonds between glass and frame

PC13. Look for any moisture or corrosion present in the PV modules

PC14. Locate any burned connections inside module (hot spots)

PC15. Identify any grounding corrosion of wires or frame

PC16. Establish any weakness or corrosion in the PV Array mount

PC17. Clean the PV panels to remove any dirt, leaves and debris accumulation.

PC18. To regularly check the proper working of batteries -electrolyte level, soiling of connectors etc
PC19. Use AC/DC clamp meter and earth tester
PC20. use infrared (IR) thermometer to check any higher temperatures (hot spots) occurrence at circuit breakers, terminals, wires etc.
PC21. Check Junction box and module connection failures
PC22. Check bypass Diode failures
PC23. Rectify any open circuiting leading to arcing
PC24. Maintain a checklist of all required maintenance tasks and their recommended intervals
PC25. utilize Condition-based maintenance (CBM) by using real-time data to prioritize and optimize maintenance and resources
PC26. Maintain a proper record of all the maintenance issues faced and rectified along with root cause analysis

Solar PV Safety

To be competent, the user/individual on the job must be able to:

PC27. Understand the Importance of Solar PV Safety
PC28. Understanding the top risks associated with Solar PV and how to avoid the same
PC29. Prevent shock and electrocution from energized conductor
PC30. to check that the solar PV system is grounded correctly and that all wiring is securely insulated
PC31. ensure that all electrical connections are secure and corrosion-free to prevent this risk.
PC32. Ensure that the system is complying to the codes & standards and that all of the equipment is in good operating order
PC33. Install Arc Fault Circuit Interrupters (AFCIs) on the inverter or the AC breaker panel to detect and interrupt an arc fault before it can cause damage or fire.
PC34. Install Ground Fault Circuit Interrupters (GFCIs) to detect and interrupt ground faults can prevent damage and fire hazards.
PC35. Ensure Proper Labeling of electrical equipment with warning labels - hazard level and the necessary personal protective equipment to be worn
PC36. to wear the appropriate safety gear, such as rubber-soled shoes and gloves, face shield, hard hat etc
PC37. understand the hazards associated with an arc flash and how to work safely around electrical equipment.
PC38. Install a DC disconnect switch can help to quickly and safely isolate the DC side of the solar PV system in the event of an arc flash.
PC39. Conduct regular risk assessments with an arc flash and implementing appropriate measures to reduce or eliminate those risks.
PC40. Implement an emergency response plan in place to minimize the damage and injuries in the event of an accident
PC41. Use proper fall protection equipment, such as harnesses and safety lines, to prevent falls from the rooftop.
PC42. Ensure all equipment and tools are in good working condition before using them.
PC43. Follow proper lifting techniques to prevent injuries when lifting and moving heavy equipment or materials.

PC44. Use lifeline while working on an industrial shed roof to prevent falls from heights
PC45. Report any near miss event
PC46. Document all the safety related issues faced and mitigation

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. safety compliances, regulatory requirements, relevant state legislations and policies on renewable energy
KU2. basic electrical and solar terms such as current, voltage, resistance, power, energy, irradiance etc. as well as signs, symbols, measurement units, graphical representation, etc
KU3. electrical safety practices in solar PV installations in terms of safe handling, protection from DC&AC disconnects, lightening, surge, grounding, etc.
KU4. safety and operational standards methods used to secure and route cables, etc
KU5. requirements such as avoidance of inductive loops, necessity of earthing systems, and other requirements
KU6. safe disposal of toxic and non-toxic wastes
KU7. basic concepts of electricity along with electrical safety
KU8. preventive and safety housekeeping measures for eliminating risks and hazard
KU9. importance of conducting safety drills and working in safe environment
KU10. the concept of Mission Karmayogi
KU11. the concept of Swachhata Hi Seva

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and interpret data from various sources
GS2. take decision with systematic course of actions and/or response
GS3. plan to manage time and equipment utilisation effectively
GS4. understand the various colour codes, as per standard electrical, mechanical and civil nomenclature
GS5. prepare and maintain proper documentation
GS6. use reasoning skills to identify and resolve basic problems
GS7. communicate effectively and share the information efficiently
GS8. Effective use of computers (MS office tools)
GS9. Presentation skills -oral and written.
GS10. Effective use read and interpret the data
GS11. read instructions, guidelines/procedures and reports

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Monitoring & Maintaining the Fuel Station Solar rooftop installation	**25**	**51**	**-**	**-**
PC1. Understand different strategies adopted for maintenance of a solar roof top PV system- Routine maintenance, Preventative maintenance, Corrective or Reactive maintenance, and Condition-based maintenance.	1	2	-	-
PC2. Understand the objective of preventive maintenance	1	1	-	-
PC3. Understand the objective of a PV system monitoring on regular basis	1	2	-	-
PC4. Track energy production and usage in real-time	-	2	-	-
PC5. Identify dips in output or problems with individual panels, batteries, or other system components.	1	2	-	-
PC6. Understand the sensors and communication devices attached to solar arrays	1	2	-	-
PC7. Read, record, & store the voltage, current, and power output in real time	1	2	-	-
PC8. Carry out routine inspection and servicing of equipment to prevent breakdowns and unnecessary production losses.	1	2	-	-
PC9. To regularly check basic maintenance elements such as modules, inverters, charge controllers, and batteries.	1	2	-	-
PC10. Carry out a thorough inspection of PV modules visually to look for any issues	1	2	-	-
PC11. Look for physical damage to frame and glass etc and repair the same	1	2	-	-
PC12. Identify any delamination or change in color of the module's outer layer due to separation of bonds between glass and frame	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Look for any moisture or corrosion present in the PV modules	1	2	-	-
PC14. Locate any burned connections inside module (hot spots)	1	2	-	-
PC15. Identify any grounding corrosion of wires or frame	1	2	-	-
PC16. Establish any weakness or corrosion in the PV Array mount	1	2	-	-
PC17. Clean the PV panels to remove any dirt, leaves and debris accumulation.	1	2	-	-
PC18. To regularly check the proper working of batteries -electrolyte level, soiling of connectors etc	1	2	-	-
PC19. Use AC/DC clamp meter and earth tester	1	2	-	-
PC20. use infrared (IR) thermometer to check any higher temperatures (hot spots) occurrence at circuit breakers, terminals, wires etc.	1	2	-	-
PC21. Check Junction box and module connection failures	1	2	-	-
PC22. Check bypass Diode failures	1	2	-	-
PC23. Rectify any open circuiting leading to arcing	1	2	-	-
PC24. Maintain a checklist of all required maintenance tasks and their recommended intervals	1	2	-	-
PC25. utilize Condition-based maintenance (CBM) by using real-time data to prioritize and optimize maintenance and resources	1	2	-	-
PC26. Maintain a proper record of all the maintenance issues faced and rectified along with root cause analysis	1	2	-	-
Solar PV Safety	**20**	**39**	**-**	**-**
PC27. Understand the Importance of Solar PV Safety	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC28. Understanding the top risks associated with Solar PV and how to avoid the same	1	2	-	-
PC29. Prevent shock and electrocution from energized conductor	1	2	-	-
PC30. to check that the solar PV system is grounded correctly and that all wiring is securely insulated	1	2	-	-
PC31. ensure that all electrical connections are secure and corrosion-free to prevent this risk.	1	2	-	-
PC32. Ensure that the system is complying to the codes & standards and that all of the equipment is in good operating order	1	2	-	-
PC33. Install Arc Fault Circuit Interrupters (AFCIs) on the inverter or the AC breaker panel to detect and interrupt an arc fault before it can cause damage or fire.	1	2	-	-
PC34. Install Ground Fault Circuit Interrupters (GFCIs) to detect and interrupt ground faults can prevent damage and fire hazards.	1	2	-	-
PC35. Ensure Proper Labeling of electrical equipment with warning labels - hazard level and the necessary personal protective equipment to be worn	1	2	-	-
PC36. to wear the appropriate safety gear, such as rubber-soled shoes and gloves, face shield, hard hat etc	1	2	-	-
PC37. understand the hazards associated with an arc flash and how to work safely around electrical equipment.	1	2	-	-
PC38. Install a DC disconnect switch can help to quickly and safely isolate the DC side of the solar PV system in the event of an arc flash.	1	2	-	-
PC39. Conduct regular risk assessments with an arc flash and implementing appropriate measures to reduce or eliminate those risks.	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC40. Implement an emergency response plan in place to minimize the damage and injuries in the event of an accident	1	2	-	-
PC41. Use proper fall protection equipment, such as harnesses and safety lines, to prevent falls from the rooftop.	1	2	-	-
PC42. Ensure all equipment and tools are in good working condition before using them.	1	2	-	-
PC43. Follow proper lifting techniques to prevent injuries when lifting and moving heavy equipment or materials.	1	2	-	-
PC44. Use lifeline while working on an industrial shed roof to prevent falls from heights	1	2	-	-
PC45. Report any near miss event	1	1	-	-
PC46. Document all the safety related issues faced and mitigation	1	2	-	-
NOS Total	**45**	**90**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N3117
NOS Name	Maintaining Solar PV System and Ensuring Safety at a Fuel Station
Sector	Hydrocarbon
Sub-Sector	
Occupation	Retail Distribution
NSQF Level	4
Credits	3.5
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

HYC/N9301: Working Effectively in a team

Description

This unit is about working effectively within a team.

Scope

The scope covers the following :

- Effective team work

Elements and Performance Criteria

Effective team work

To be competent, the user/individual on the job must be able to:

PC1. maintain clear communication with colleagues
PC2. pass on information to colleagues in line with organisational requirements
PC3. provide support to the team members
PC4. respect the colleagues
PC5. fulfil commitments made to colleagues
PC6. inform team members timely, if timelines can't be met
PC7. take the necessary initiatives to resolve the issues while working in team
PC8. adopt gender neutral behaviour while interacting with colleagues
PC9. offer assistance to a person with disability (PWD), only if required

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the organization policies and procedures related to team performance
KU2. the importance of effective communication and establishing good working relationships with colleagues
KU3. the importance of creating an environment of trust and mutual respect
KU4. the implications of own work on the work and schedule of others
KU5. the standard practices in organisation w.r.t communication at various levels
KU6. the personal responsibility for completing the task in time
KU7. importance of gender equality
KU8. importance of showing empathy while interacting with a PwD

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. communicate effectively in writing
GS2. read instructions, guidelines/procedures
GS3. work in a disciplined manner for meeting commitments and deadline
GS4. how to plan and prioritise the work
GS5. the importance of consistent and reliable services
GS6. apply problem solving approaches in different situations
GS7. apply balanced judgments to different situations

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Effective team work	**20**	**30**	**-**	**-**
PC1. maintain clear communication with colleagues	2	3	-	-
PC2. pass on information to colleagues in line with organisational requirements	2	3	-	-
PC3. provide support to the team members	2	4	-	-
PC4. respect the colleagues	3	4	-	-
PC5. fulfil commitments made to colleagues	2	3	-	-
PC6. inform team members timely, if timelines can't be met	2	4	-	-
PC7. take the necessary initiatives to resolve the issues while working in team	3	4	-	-
PC8. adopt gender neutral behaviour while interacting with colleagues	2	2	-	-
PC9. offer assistance to a person with disability (PWD), only if required	2	3	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N9301
NOS Name	Working Effectively in a team
Sector	Hydrocarbon
Sub-Sector	Generic
Occupation	Generic, Generic
NSQF Level	4
Credits	2
Version	3.0
Last Reviewed Date	16/04/2026
Next Review Date	16/04/2029
NSQC Clearance Date	16/04/2026

HYC/N9302: Maintain health, safety and security procedures

Description

This unit is about maintaining health, safety and security procedure at workplace. It covers responsibilities towards self, others, assets and the environment.

Scope

The scope covers the following :

- Follow health and safety measures
- Follow safety procedures during emergency

Elements and Performance Criteria

Follow health and safety measures

To be competent, the user/individual on the job must be able to:

PC1. use protective clothing/equipment such as face mask, hand gloves, goggle etc for specific tasks and work conditions

PC2. identify the people responsible for maintaining health and safety in the workplace

PC3. identify possible causes of risk or accident in the workplace

PC4. follow safe working practices while dealing with hazards to ensure the safety of self and others

PC5. lift heavy objects safely using correct procedures

PC6. follow safety signages

PC7. maintain hands hygiene by washing hand frequently and thoroughly with soap and water or alcohol-based hand rub

PC8. inform the concerned person of any illness related to self and others

PC9. maintain workplace hygiene by disinfecting the equipment and tools regularly

Follow safety procedures during emergency

To be competent, the user/individual on the job must be able to:

PC10. respond promptly and appropriately to an accident or in an emergency situation

PC11. use appropriate fire extinguishers for different types of fires correctly

PC12. follow appropriate rescue techniques during fire hazard

PC13. follow good housekeeping practice in order to prevent fire hazards

PC14. inform fire safety department about any near-miss incidents in the work place

PC15. provide appropriate first aid to victims in an emergency situation

PC16. follow the applicable regulations and codes as per safety standard

PC17. prepare written accident/incident report and share with the concerned officer/department

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. company's policies on personnel management and duty reporting procedure
KU2. reporting structure within organization
KU3. health and safety hazards commonly affecting the work environment and related precautions
KU4. importance of maintaining personal hygiene using PPE kit, sanitizer and soap
KU5. importance of maintaining workplace hygiene
KU6. preventative and remedial actions to be taken in the case of exposure to toxic materials
KU7. importance of using protective clothing/equipment while working
KU8. various causes of fire
KU9. techniques of using different types of fire extinguishers
KU10. different materials used for extinguishing fire
KU11. various types of safety signs and their significance

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. communicate effectively by writing
GS2. read instructions, guidelines/procedures and reports
GS3. identify and report potential sources of danger
GS4. how to plan the work to meet the deadline
GS5. the importance of on time services
GS6. apply problem solving approaches in different situations
GS7. apply balanced judgments in different situations

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Follow health and safety measures	**9**	**15**	**-**	**-**
PC1. use protective clothing/equipment such as face mask, hand gloves, goggle etc for specific tasks and work conditions	1	2	-	-
PC2. identify the people responsible for maintaining health and safety in the workplace	1	-	-	-
PC3. identify possible causes of risk or accident in the workplace	1	2	-	-
PC4. follow safe working practices while dealing with hazards to ensure the safety of self and others	1	2	-	-
PC5. lift heavy objects safely using correct procedures	1	2	-	-
PC6. follow safety signages	1	2	-	-
PC7. maintain hands hygiene by washing hand frequently and thoroughly with soap and water or alcohol-based hand rub	1	2	-	-
PC8. inform the concerned person of any illness related to self and others	1	1	-	-
PC9. maintain workplace hygiene by disinfecting the equipment and tools regularly	1	2	-	-
Follow safety procedures during emergency	**11**	**15**	**-**	**-**
PC10. respond promptly and appropriately to an accident or in an emergency situation	1	2	-	-
PC11. use appropriate fire extinguishers for different types of fires correctly	2	2	-	-
PC12. follow appropriate rescue techniques during fire hazard	1	2	-	-
PC13. follow good housekeeping practice in order to prevent fire hazards	1	1	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. inform fire safety department about any near-miss incidents in the work place	2	2	-	-
PC15. provide appropriate first aid to victims in an emergency situation	1	2	-	-
PC16. follow the applicable regulations and codes as per safety standard	1	2	-	-
PC17. prepare written accident/incident report and share with the concerned officer/department	2	2	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N9302
NOS Name	Maintain health, safety and security procedures
Sector	Hydrocarbon
Sub-Sector	Generic
Occupation	Generic, Generic
NSQF Level	4
Credits	2
Version	3.0
Last Reviewed Date	16/04/2026
Next Review Date	16/04/2029
NSQC Clearance Date	16/04/2026

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. identify employability skills required for jobs in various industries

PC2. identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.

PC4. follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC5. recognize the significance of 21st Century Skills for employment

PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC7. use basic English for everyday conversation in different contexts, in person and over the telephone

PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English

PC9. write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

PC10. understand the difference between job and career

PC11. prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings

PC13. work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC14. communicate and behave appropriately with all genders and PwD

PC15. escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC16. select financial institutions, products and services as per requirement

PC17. carry out offline and online financial transactions, safely and securely

PC18. identify common components of salary and compute income, expenses, taxes, investments etc

PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC20. operate digital devices and carry out basic internet operations securely and safely

PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively

PC22. use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research

PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion

PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

PC26. identify different types of customers

PC27. identify and respond to customer requests and needs in a professional manner.

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

GS3. behave politely and appropriately with all
GS4. how to work in a virtual mode
GS5. perform calculations efficiently
GS6. solve problems effectively
GS7. pay attention to details
GS8. manage time efficiently
GS9. maintain hygiene and sanitization to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	**1**	**1**	**-**	**-**
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
Constitutional values – Citizenship	**1**	**1**	**-**	**-**
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	**2**	**4**	**-**	**-**
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
Basic English Skills	**2**	**3**	**-**	**-**
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
Career Development & Goal Setting	**1**	**2**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
Communication Skills	**2**	**2**	**-**	**-**
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**2**	**-**	**-**
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
Financial and Legal Literacy	**2**	**3**	**-**	**-**
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
Essential Digital Skills	**3**	**4**	**-**	**-**
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Entrepreneurship	**2**	**3**	**-**	**-**
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
Customer Service	**1**	**2**	**-**	**-**
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**2**	**3**	**-**	**-**
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.

2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.

3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.

4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training centre (as per assessment criteria below).

5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.

6. To pass the Qualification Pack assessment, every trainee should score a minimum of 70% of aggregate marks to successfully clear the assessment.

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 70

(**Please note**: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HYC/N3115.Introduction to Solar Technologies and Role & Responsibilities of a Solar Panel Installer at a Retail Outlet	75	150	0	0	225	20
HYC/N3116.Assembling and Commissioning of Rooftop Solar PV System at a Fuel Station	55	110	0	0	165	20
HYC/N3117.Maintaining Solar PV System and Ensuring Safety at a Fuel Station	45	90	0	0	135	20
HYC/N9301.Working Effectively in a team	20	30	-	-	50	15
HYC/N9302.Maintain health, safety and security procedures	20	30	-	-	50	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	**235**	**440**	**-**	**-**	**675**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
OS	Occupational Standard(s)
KU	Knowledge and Understanding
GS	Generic Skills
FAQ	Frequently Asked Questions
BP	Business Partner
KYC	Know Your Consumer
FAB	Feature Advantage Benefit
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
OS	Occupational Standard(s)
KU	Knowledge and Understanding
GS	Generic Skills
FAB	Frequently Asked Questions
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OS	Occupational Standard(s)
KU	Knowledge and Understanding
GS	Generic Skills
FAB	Frequently Asked Questions
BP	Business Partner
KYC	Know Your Consumer
FAB	Feature Advantage Benefit
SS	Stainless Steel
PPE	Personal Protective Equipment

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.